Appendix D.2 — Functional and Non-Functional Requirements Specification

Purpose

This document presents the complete requirements specification supporting §3.4 of the dissertation. Table D.2 identifies the 12 functional and seven non-functional requirements, their corresponding research objectives and the design areas responsible for addressing them.

ID	Requirement	Objective Addressed	Design Area
Functional			
FR01	Guest and registered students can ask conceptual Scala questions and continue multi-turn chats.	2	Chat and context (§4.6, §4.9)
FR02	Retrieve indexed course evidence and assess support for the whole question.	2	Retrieval and evidence assessment (§4.6)
FR03	Generate supported Tier-1 answer with tier labels, citations and source-location details.	2, 3	Grounded generation and citations (§4.6, §4.9)
FR04	When course support is insufficient, attempt labelled Tier-2 generation through OpenAI API without course citations.	2, 3	External fallback (§4.7, §4.9)

FR05	Route insufficient evidence and retrieval failures to controlled fallback; return controlled errors for non-recoverable failures.	2	Failure handling (§4.6-§4.7)
FR06	Students can register, log in/out, reset passwords, change email/password, and delete their accounts.	2	Account management (§4.8, §4.10)
FR07	Registered students can retain, view, reopen, rename, delete and archive their own conversation history.	2	Private conversation history (§4.8, §4.10)
FR08	Administrator can add supported sources, monitor ingestion and retry failures; only successfully indexed sources enter retrieval	2	Source ingestion and administration (§4.5, §4.9)
FR09	Administrators can delete course sources, retaining historical citation snapshots.	2	Source deletion and citation preservation (§4.8-§4.9)
FR10	Administrator can view user status and aggregate source usage by registered users and conversations, excluding guests.	2	Administrator reporting (§4.8-§4.9)
FR11	Administrator can configure the OpenAI generation models independently for Tier 1 and Tier 2.	2	Model configuration (§4.8-§4.9)

FR12	Administrators can log in/out and change their email address and password.	2	Administrator account security (§4.9-§4.10)
Non-functional			
NFR01	Usability: Provide clear, consistent interactions without requiring students to understand RAG or LLM architecture.	2,4	Interface and feedback (§4.9)
NFR02	Transparency: Identify response origins through tier labels and traceable citation information.	2,3	Response provenance (§4.6- §4.7, §4.9)
NFR03	Reliability: Isolate ingestion failures, continue queued work and exclude incomplete or failed sources from retrieval.	2	Queue and source eligibility (§4.5)
NFR04	Security/Privacy: Encrypt stored conversation, enforce owner-only access, prevent in-app administrator access to student conversations and protect backend secrets.	2	Encryption and access (§4.8, §4.10)
NFR05	Maintainability: Separate responsibilities so components can be tested or modified with limited impact elsewhere.	2	Modular architecture (§4.4)
NFR06	Extensibility: Allow approved materials to be added or replaced without redesigning the QA architecture.	2	Knowledge-base design (§4.5)

NFR07	Model settings persist across restarts and remain consistent throughout each generation's request.	2	Configuration and request consistency (§4.6-§4.8)

Table D.2: Functional and Non-Functional Requirements and Traceability to Objectives and Design

The implementation and verification of these requirements are documented in Chapter 5, with detailed requirement-to-test mappings provided in Appendix D.3.